# The Solutions 2021

B—bookwork, A—application, E—new example, T—new theory

1.

a)

i)

We have

| $X_1X_2$ | 00 | 01 | 10 | 11 |
|---|---|---|---|---|
| y | 0 | 0 | 0 | 1 |

$$P(y=0) = p^2 + 2pq = 0.36$$

$$P(y=1) = q^2 = 0.64$$

$H(y) = H(0.36) = 0.943$

[3A]

ii) We have the conditional distribution P(y|$x_1$):

| | y=0 | y=1 |
|---|---|---|
| $x_1$=0 | 1 | 0 |
| $x_1$=1 | p | q |

$$H(y|x_1) = 0 + pH(p) = 0.2 \times H(0.2) = 0.144$$

$$I(x_1; y) = H(y) - H(y|x_1) = 0.943 - 0.144 = 0.799$$

[4A]

iii) $I(x_{1,2}; y) = H(y) - H(y|x_{1,2}) = H(y) - 0$ since y is a function of $x_{1,2}$.

$$I(x_{1,2}; y) = H(y) = 0.943$$

[3A]

b)

i) The entropy rate of $x_n$ is given by

$H(x_n) = \frac{1}{2}\log(2\pi e\sigma_x^2) = \frac{1}{2}\log(2\pi e)$. [3A]

Obviously, $y_n$ has zero mean, while its variance is

$$\sigma_y^2 = \sum_{i=0}^{\infty} 4^{-i} = \frac{4}{3}$$

so that

$$H(y_n) = \frac{1}{2}\log(2\pi e\sigma_y^2) = \frac{1}{2}\log(2\pi e \cdot \frac{4}{3})$$

What about the joint entropy? We need to find the covariance matrix.

Note that the correlation is given by

$$E[x_n y_n] = 1$$

Thus the covariance matrix is [4A]

$$K = \begin{bmatrix} 1 & 1 \\ 1 & \frac{4}{3} \end{bmatrix}$$

and

$$H(x_n, y_n) = \frac{1}{2}\log|2\pi eK|$$

It follows that the entropy rate is given by

$$I(x_n; y_n) = H(x_n) + H(y_n) - H(x_n, y_n)$$
$$= \tfrac{1}{2}\log\left(\tfrac{\sigma_x^2\sigma_y^2}{|K|}\right) = \tfrac{1}{2}\log\left(\tfrac{1\cdot\frac{4}{3}}{\frac{1}{3}}\right) = 1 \text{ bit}$$

[3A]

ii) Let's calculate the conditional entropy [2A]

$$h(y_n|y_{1:n-1}) = h(y_n|y_{n-1})$$
$$= h\left(x_n + \frac{1}{2}x_{n-1} + \frac{1}{4}x_{n-2} + \cdots \middle| x_{n-1} + \frac{1}{2}x_{n-2} + \frac{1}{4}x_{n-3} + \cdots\right)$$
$$= h\left(x_n + \frac{1}{2}y_{n-1}\middle| y_{n-1}\right)$$
$$= h(x_n|y_{n-1}) = h(x_n)$$

Therefore, the entropy rate

$$h(y) = h(x_n) = \tfrac{1}{2}\log(2\pi e) = 2.047 \text{ bits}$$

[3A]

2.

b) a) 0100011011000001010011…

| location | parsing | encoding |
|---|---|---|
| 0 | | |
| 1 | 0 | (0,0) |
| 10 | 1 | (0,1) |
| 11 | 00 | (1,0) |
| 100 | 01 | (1,1) |
| 101 | 10 | (10,0) |
| 110 | 11 | (10,1) |
| 111 | 000 | (11,0) |
| 1000 | 001 | (11,1) |
| 1001 | 010 | (100,0) |
| 1010 | 011 | (100,1) |
| 1011 | 100 | |
| 1100 | 101 | |
| 1101 | | |
| 1110 | | |
| 1111 | | |

[5E, 0.5 each step]

iii) Basically, it is impossible to compress this sequence, so the average number of bits needed is 1.

[5E]

b)

i)

Joint distribution (row indexed by x, column indexed by y) [2A]

$$\begin{bmatrix} 0.15 & 0.075 & 0.075 \\ 0.125 & 0.25 & 0.125 \\ 0.05 & 0.05 & 0.1 \end{bmatrix}$$

The MAP estimator picks the maximum entry in each column:

$$\hat{x} = \begin{cases} 0 & if\ y = 0 \\ 1 & if\ \ y = 1 \\ 1 & if\ \ y = 2 \end{cases}$$

Error probability

$P_e = 1 - (0.15 + 0.25 + 0.125) = 0.475$ [3A]

ii)

Marginal distribution p(y = 0) = 0.325, p(y = 1) = 0.375, p(y = 2) = 0.3.

Conditional distribution p(x|y) (row indexed by x, column indexed by y) [2A]

$$\begin{bmatrix} \frac{0.15}{0.325} & \frac{0.075}{0.375} & \frac{0.075}{0.3} \\ \frac{0.125}{0.325} & \frac{0.25}{0.375} & \frac{0.125}{0.3} \\ \frac{0.05}{0.325} & \frac{0.05}{0.375} & \frac{0.1}{0.3} \end{bmatrix} = \begin{bmatrix} 0.462 & 0.2 & 0.25 \\ 0.385 & 0.667 & 0.417 \\ 0.154 & 0.133 & 0.333 \end{bmatrix}$$

Since H(x|y) is the average column entropy, we have [3A]

$$H(y|x) = 0.325 \times H(0.462,0.385,0.154) + 0.375 \times H(0.2,0.667,0.133) + 0.3 \times H(0.25,0.417,0.333)$$

$$= 0.325 \times 1.461 + 0.375 \times 1.241 + 0.3 \times 1.555$$
$$= 1.407$$

iii) By Fano's inequality

$$p_e \geq \frac{H(x \mid y) - 1}{\log(|X| - 1)} = 0.407$$

which is smaller than 0.475, meaning it's a valid lower bound. [5A]

3.

a)

The capacity is given by [3A]

$$C = W_1 \log\left(1 + \frac{P_1}{N\ W_1}\right) + W_2 \log\left(1 + \frac{P_2}{N\ W_2}\right)$$
$$= W_1 \log\left(1 + \frac{P_1}{N\ W_1}\right) + (W - W_1) \log\left(1 + \frac{P_2}{N\ (W - W_1)}\right)$$

Take the derivative (use the natural log for convenience): [3T]

$$\log\left(1 + \frac{P_1}{NW_1}\right) + \frac{W_1}{1 + \frac{P_1}{NW_1}}\left(-\frac{P_1}{NW_1^2}\right)$$
$$-\log\left(1 + \frac{P_2}{N(W - W_1)}\right) + \frac{W - W_1}{1 + \frac{P_2}{N(W - W_1)}}\left(\frac{P_2}{N(W - W_1)^2}\right) = 0$$

We obtain [4T]

$$W_1 = \frac{P_1}{P_1 + P_2} W$$

b) The joint distribution is given by: [3E]

| $z_1$ | $z_2$ | x | y | probability |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | $p^2$ |
| 0 | 1 | 0 | -1 | pq |
| 1 | 0 | 0 | 1 | pq |
| 1 | 1 | 1 | 0 | $q^2$ |

From the table, [3E]

$$H(y|x) = H(y|x = 0)P(x = 0) + H(y|x = 1)P(x = 1)$$
$$= H(y|x = 0)P(x = 0)$$
$$= H(\frac{p^2}{p^2 + 2pq}, \frac{pq}{p^2 + 2pq}, \frac{pq}{p^2 + 2pq})(p^2 + 2pq)$$

And [3E]

$$H(x|y) = H(x|y = 0)P(y = 0) + H(x|y = 1)P(y = 1) + H(x|y = -1)P(y = -1)$$
$$= H(x|y = 0)P(y = 0)$$
$$= H(\frac{p^2}{p^2 + q^2}, \frac{q^2}{p^2 + q^2})(p^2 + q^2)$$

Also

$$H(x) = H(p^2 + 2pq, q^2)$$

So the joint entropy [3E]

$$H(x,y) = H(p^2 + 2pq, q^2) + H\left(\frac{p^2}{p^2+2pq}, \frac{pq}{p^2+2pq}, \frac{pq}{p^2+2pq}\right)(p^2 + 2pq)$$
$$= 2H(p)$$

Slepian-Wolf region is

$$R_X \geq H(X|Y)$$
$$R_Y \geq H(Y|X)$$
$$R_X + R_Y \geq H(X,Y)$$

[3E]

4. a)

$$H(X) = H(Y) = 1$$

$$H(X,Y) = H(X) + H(Y|X) = H(X) + H(p)$$
$$= 1 - 0.9 \log 0.9 - 0.1 \log 0.1 = 1.469$$

$$I(X;Y) = H(Y) - H(Y|X) = 0.531$$

[5A]

b)

Since $X$ is uniformly distributed, every sequence has probability $2^{-n}$, hence for every sequence,

$$-\frac{1}{n} \log p(X) = 1 = H(X)$$

Therefore, all the $2^n$ sequences are in $T_\varepsilon^{(n)}(X)$.

The same is true for y.

[5A]

c)

$(X,Y)$ is jointly typical

$\Rightarrow$ both X and Y are typical (from the first two conditions)

The third condition

$$-\frac{1}{n}\log p(x,y) = -\frac{1}{n}\log\left[2^{-n}p^{k}(1-p)^{n-k}\right]$$

$$= 1 - \frac{k}{n}\log p - \frac{n-k}{n}\log(1-p)$$

$(X,Y)$ is jointly typical $\Rightarrow$

$$\left|1 - \frac{k}{n}\log p - \frac{n-k}{n}\log(1-p) - H(X,Y)\right| < \varepsilon$$

$$\left|-\frac{k}{n}\log p - \frac{n-k}{n}\log(1-p) - H(p)\right| < \varepsilon$$

$\Rightarrow$ z is typical

[5A]

d)

Since every x is typical, we only need to be concerned with z. Notice that $H(z) = H(p) = 0.469$.

For ε = 0.1, the table shows that the following rows are typical (0.369 – 0.569):

k = 2, 3.

Size = 300 + 2300 = 2600

Probability of joint typicality = 0.2265 + 0.1384 = 0.365

[5A]

e)

Probability of error = Probability(not jointly typical or jointly typical with another codeword)

<= Prob(not jointly typical) + Prob(jointly typical with any other codewords)

Let's consider them case by case.

Prob(not jointly typical) = 1 – 0.365 = 0.635

Prob(jointly typical with any other codewords) <= $(2^{nR} - 1)$ x Prob(jointly typical with another codeword)

Since the received vector y is independent from another codeword x', their joint distribution is

$$P(\boldsymbol{x}', \boldsymbol{y}) = 2^{-n}, \quad \forall \boldsymbol{x}', \boldsymbol{y}$$

meaning that they cannot be jointly typical.

All in all,

Probability of error = Prob(not jointly typical) + 0

=0.635

[5T]